

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Chemistry — SSC-I (9th Grade)

Syllabus: Chapter 13 — Organic Chemistry

ANSWER KEY (Version 1)

Total Marks: 65

Time Allowed: 2h 30m

Version: 1

SECTION A — Multiple Choice Questions Answer Key (12 × 1 = 12 Marks)

Q1. Organic compounds are defined as hydrocarbons and their: ✓ **B — derivatives** 1 mark

The textbook (section 13.1) states: "Organic compounds are also defined as the **hydrocarbons and their derivatives**." Allotropes and isotopes relate to elements, not compound classes. Isomers are a structural phenomenon, not the defining scope of organic chemistry.

Q2. Adjacent members of a homologous series differ by: ✓ **B — $-\text{CH}_2-$** 1 mark

Section 13.2 defines a homologous series as one in which "any two adjacent molecules differ by $-\text{CH}_2-$ group." The methyl group ($-\text{CH}_3$) has 3 H atoms; $-\text{C}_2\text{H}_4-$ is double the unit; $-\text{OH}$ is a functional group, not the repeating structural difference.

Q3. General formula of alkanes: ✓ **B — $\text{C}_n\text{H}_{2n+2}$** 1 mark

Table 13.1 lists alkane general formula as $\text{C}_n\text{H}_{2n+2}$. Option A (C_nH_{2n}) is for alkenes. Option C ($\text{C}_n\text{H}_{2n-2}$) is for alkynes. Option D ($\text{C}_n\text{H}_{2n+1}\text{OH}$) is for alcohols. A common error is confusing alkane and alkene formulae.

Q4. Same molecular formula C_4H_{10} , different structures: ✓ **B — isomerism** 1 mark

Section 13.5: "The compounds that have the same molecular formula but different arrangement of atoms in their molecules are called **structural isomers**. This phenomenon is called isomerism." Allotropy is for elements; homology means successive members of the same series; catenation is carbon's chain-forming ability.

Q5. $\text{CH}_3-\text{CH}=\text{CH}_2$ and $\text{CH}_2=\text{CH}-\text{CH}_3$ (same formula, double bond at same position): ✓ **B — positional isomerism** 1 mark

These two are in fact the same compound written in reverse (propene). The classic textbook example of positional isomers from section 13.5 is **1-butene** ($\text{CH}_3-\text{CH}_2-\text{CH}=\text{CH}_2$) and **2-butene** ($\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3$): same molecular formula C_4H_8 , double bond at different positions. This type of structural isomerism where the functional group shifts position is called **positional isomerism**. Chain isomerism involves a different carbon skeleton.

Q6. $-\text{OH}$ attached to an alkyl chain: ✓ **C — alcohol** 1 mark

Section 13.7.2: "Alcohols are characterized by the presence of the hydroxyl group **$-\text{OH}$ attached to a hydrocarbon chain**." If $-\text{OH}$ is attached to a *benzene ring* instead, the compound is a phenol (section 13.7.3). Ethers have oxygen between two alkyl groups; aldehydes have $-\text{CHO}$.

Q7. Carbonyl carbon bonded to one H ($-\text{CHO}$) identifies: ✓ **D — aldehyde** 1 mark

Section 13.7.5: "An aldehyde has at least one hydrogen atom attached to the carbonyl carbon atom." The condensed form is written as **$-\text{CHO}$** . A ketone has two alkyl groups on the carbonyl carbon (no H). An ester has $-\text{C}(=\text{O})-\text{O}-$; a carboxylic acid has $-\text{COOH}$. Students frequently confuse aldehyde ($-\text{CHO}$) with ketone ($-\text{CO}-$).

Q8. Carbonyl carbon in a ketone is bonded to: ✓ **C — two alkyl groups** 1 mark

Section 13.7.5: "A ketone has **two hydrocarbon groups (alkyl) bonded to the carbonyl carbon atom.**" This distinguishes ketones from aldehydes (one H on carbonyl C). Propanone (acetone, CH_3COCH_3) is the textbook example. Option A describes an aldehyde; option B describes a carboxylic acid functional arrangement.

Q9. $\text{CH}_3\text{—O—CH}_3$ belongs to: ✓ **C — ether** 1 mark

Section 13.7.4: "Organic compounds that have two alkyl groups attached to the same oxygen atom are called **ethers**. These compounds have C—O—C linkage." $\text{CH}_3\text{—O—CH}_3$ is named dimethyl ether in the textbook. A common error is confusing this C—O—C pattern with an alcohol (R—OH) or ester (R—C(=O)—O—R').

Q10. Functional group of carboxylic acids: ✓ **C — —COOH** 1 mark

Section 13.7.6: "The functional group of organic acid is called the **carboxyl group**, written as —COOH." This combines a carbonyl (—C=O) with a hydroxyl (—OH) on the same carbon. Option A (—CHO) is the aldehyde group. Option B (—CO—) is the ketone carbonyl. Option D (—COO—) is part of the ester linkage but lacks the H.

Q11. Which compound is an ester? ✓ **C — $\text{CH}_3\text{COOCH}_3$** 1 mark

Section 13.7.7: esters have general formula R—C(=O)—O—R'. $\text{CH}_3\text{COOCH}_3$ is **methyl ethanoate (methyl acetate)**, given as the first textbook example. Option A (CH_3COOH) is ethanoic acid (carboxylic acid). Option B (CH_3COCH_3) is propanone (ketone). Option D (CH_3CHO) is ethanal (aldehyde). Confusing ester (—COO—) with ether (—O—) or carboxylic acid (—COOH) is a very common error.

Q12. Carbon's ability to bond with other carbons to form chains: ✓ **B — catenation** 1 mark

Section 13.5: "This self-linking ability of carbon is called **catenation**. There appears to be almost no limit to the number of different structures that carbon can form." Isomerism refers to same-formula different-structure compounds; homology describes the series pattern; saturation refers to bond type, not chain formation.

SECTION B — Short Questions Answer Key (11 × 3 = 33 Marks)

2(i) Organic chemistry definition + exclusions 3 marks

Definition 1 mark: The study of carbon-containing compounds and their properties is called **organic chemistry**.

Why CO_2 , CO, and carbonates are excluded 2 marks: Although these compounds contain carbon, they are classified as **inorganic substances** because they have completely different properties from organic compounds. They lack the carbon-hydrogen (C—H) backbone that characterises organic molecules. The textbook (p. 174) specifically lists CO_2 , CO, carbonates and carbides as exceptions.

OR

Organic compounds defined 1 mark: Carbon-containing compounds (hydrocarbons and their derivatives) whose essential constituent is carbon in combination with other elements.

Four common elements besides carbon 2 marks: **Hydrogen (H), Oxygen (O), Nitrogen (N), Sulphur (S)** (also Phosphorus and halogens are acceptable, as stated in section 13.1).

2(ii) Homologous series definition + characteristics 3 marks

Definition 1 mark: A series of related compounds in which any two adjacent molecules differ by a $-\text{CH}_2-$ unit is called a **homologous series**.

Four general characteristics 2 marks (1 mark for any two, 2 marks for all four):

- Family members have the **same functional group**.
- Family members have the **same general formula**.
- Adjacent members differ by a $-\text{CH}_2-$ unit.
- Family members **display a trend** in physical properties.
- Family members possess **similar chemical properties**.

OR

General formulae and n=3 members 3 marks:

Series	General formula	n=3 member
Alkanes	$\text{C}_n\text{H}_{2n+2}$	C_3H_8 (propane)
Alkenes	C_nH_{2n}	C_3H_6 (propene)
Alkynes	$\text{C}_n\text{H}_{2n-2}$	C_3H_4 (propyne)
Alcohols	$\text{C}_n\text{H}_{2n+1}\text{OH}$	$\text{C}_3\text{H}_7\text{OH}$ (propanol)

2(iii) Structural formula — propane 3 marks

Definition 1 mark: A formula that describes the **arrangement of atoms in a molecule** is called a structural formula.

Propane (C_3H_8) 2 marks:

Structural formula (full): $\text{H H H} \mid \mid \mid \text{H} - \text{C} - \text{C} - \text{C} - \text{H} \mid \mid \mid$ Condensed structural formula:
 $\text{CH}_3 - \text{CH}_2 - \text{CH}_3$

OR

Butane (C_4H_{10}) 3 marks:

Molecular formula: C_4H_{10} Structural formula: $\text{H H H H} \mid \mid \mid \mid \text{H} - \text{C} - \text{C} - \text{C} - \text{C} - \text{H} \mid \mid \mid \mid$
Condensed structural formula: $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$

2(iv) Saturated vs unsaturated hydrocarbons 3 marks

Saturated 1 mark: Hydrocarbons whose **carbon-carbon bonds are all single bonds** are called saturated. Examples: methane (CH_4), ethane (CH_3CH_3).

Unsaturated 1 mark: Hydrocarbons containing **one or more carbon-carbon multiple bonds** (double or triple) are called unsaturated. Examples: ethene ($\text{CH}_2=\text{CH}_2$), ethyne ($\text{HC}\equiv\text{CH}$).

Unsaturated types 1 mark: Alkenes have at least one $\text{C}=\text{C}$ double bond (general formula C_nH_{2n}); alkynes have at least one $\text{C}\equiv\text{C}$ triple bond (general formula $\text{C}_n\text{H}_{2n-2}$).

OR

Identification with reasons 3 marks (1 mark for any two correct, max 3):

- (a) $\text{CH}_3 - \text{CH}_2 - \text{CH}_3$ — **saturated**: all $\text{C}-\text{C}$ single bonds.
- (b) $\text{CH}_3 - \text{C}\equiv\text{CH}$ — **unsaturated**: contains a $\text{C}\equiv\text{C}$ triple bond.
- (c) $\text{CH}_3 - \text{CH}=\text{CH}_2$ — **unsaturated**: contains a $\text{C}=\text{C}$ double bond.
- (d) $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ — **unsaturated**: contains two $\text{C}=\text{C}$ double bonds.

2(v) Structural isomers of C₄H₁₀ 3 marks

Definition 1 mark: Compounds that have the **same molecular formula but different arrangement of atoms** in their molecules are called **structural isomers**.

Two isomers of C₄H₁₀ 2 marks:

n-Butane: CH₃-CH₂-CH₂-CH₃ (straight chain) iso-Butane: CH₃ | CH₃-CH-CH₃ (branched chain)

OR

Positional isomers of C₄H₈ 3 marks:

1-Butene: CH₃-CH₂-CH=CH₂ (double bond at C1-C2) 2-Butene: CH₃-CH=CH-CH₃ (double bond at C2-C3)

Both have molecular formula C₄H₈ but differ in the **position** of the double bond. (Ref: section 13.5, textbook example.)

2(vi) Open-chain vs closed-chain compounds 3 marks

Open-chain (acyclic) 1 mark: Compounds that contain an open chain of carbon atoms — carbon atoms linked in a linear (straight or branched) pattern. Examples: propane (CH₃CH₂CH₃), iso-butane.

Closed-chain (cyclic) 1 mark: Organic compounds which contain **rings of atoms** are called cyclic compounds. Examples: cyclopropane, benzene.

Subcategories of open-chain 1 mark: Open-chain compounds may be **straight-chain** (all carbons in one continuous chain) or **branched-chain** (carbon side-groups attached to the main chain).

OR

Homocyclic vs heterocyclic 3 marks:

- **Homocyclic (carbocyclic):** rings contain only carbon atoms. Example: **benzene** (C₆H₆), cyclopropane. 1 mark
- **Heterocyclic:** rings contain one or more atoms *other than* carbon (e.g. N, O, S). Examples from textbook: **pyridine** (N in ring), **furan** (O in ring), **thiophene** (S in ring). 2 marks

2(vii) Functional group definition and importance 3 marks

Definition 1 mark: "An atom or group of atoms that gives a family of organic compounds its characteristic chemical and physical properties is called a **functional group**." (Section 13.7)

Three reasons for importance 2 marks (1 mark for any two):

1. Functional groups serve as the **basis for naming** organic compounds.
2. They are used to **classify organic compounds** into different classes (all compounds with the same functional group belong to the same class).
3. A functional group is the **site of chemical reactivity** in a molecule — it determines what reactions the compound undergoes.

OR

Functional group determines properties 3 marks: The properties of organic compounds are predominantly determined by the functional group present. For example, **ethanol** (CH₃CH₂OH) and **dimethyl ether** (CH₃-O-CH₃) both have molecular formula C₂H₆O, but ethanol has the **-OH (hydroxyl)** group so it acts as an alcohol (miscible with water, higher boiling point), while dimethyl ether has the **C-O-C** linkage so it is an ether (used as anaesthetic, more volatile). Same formula, completely different properties — caused by different functional groups.

2(viii) Alcohols vs ethers 3 marks

Feature	Alcohol	Ether
Functional group	–OH (hydroxyl)	C–O–C (ether linkage)
General formula	R–OH	R–O–R'
Named example	Methanol (CH ₃ OH)	Dimethyl ether (CH ₃ –O–CH ₃)

Award 1 mark per correct row (functional group, formula, example).

OR

Phenol definition 1 mark: When the –OH group is attached to a **benzene ring**, the compound is called a phenol (e.g. C₆H₅OH). In an alcohol, –OH is attached to an alkyl chain.

Classification 2 marks:

- (a) CH₃CH₂OCH₃ — **ether** (oxygen between two alkyl groups, C–O–C).
- (b) C₆H₅OH — **phenol** (–OH on benzene ring).
- (c) CH₃OH — **alcohol** (–OH on alkyl group, R–OH).

2(ix) Aldehydes vs ketones 3 marks

Feature	Aldehyde	Ketone
Functional group	–CHO (carbonyl + one H)	–CO– (carbonyl between two alkyls)
Named example	Methanal / formaldehyde (HCHO)	Propanone / acetone (CH ₃ COCH ₃)
Condensed form	R–CHO	R–CO–R' (written RCOR')

Award 1 mark per row.

OR

Identification 3 marks (1 mark each):

- (a) CH₃COCH₃ (propanone) — **ketone**: carbonyl C is between two methyl groups (–CO–).
- (b) CH₃CH₂CHO (propanal) — **aldehyde**: –CHO group at end of chain.
- (c) CH₃COOH (ethanoic acid) — **carboxylic acid**: –COOH (carboxyl group) present.

2(x) Carboxylic acids + daily-life examples 3 marks

Definition 1 mark: The functional group of organic acids is called the **carboxyl group (–COOH)**. The general formula is R–COOH.

Methanoic acid (HCOOH) 1 mark: Also called formic acid; structural formula H–C(=O)–OH. Found in **ant stings and stinging nettles**.

Ethanoic acid (CH₃COOH) 1 mark: Also called acetic acid; structural formula CH₃–C(=O)–OH. Present in **vinegar** (the textbook states it is used to flavour food and make polyvinyl acetate).

OR

Esters definition 1 mark: Compounds having general formula R–C(=O)–O–R' are called **esters**. R and R' are alkyl groups which may be the same or different.

Two named examples 2 marks:

- **Methyl ethanoate** (methyl acetate): CH₃–C(=O)–OCH₃
- **Ethyl ethanoate** (ethyl acetate): CH₃–C(=O)–OCH₂CH₃

2(xi) Amines and amides 3 marks

Amines 1.5 marks: Functional group -NH_2 ; general formula R-NH_2 . Example: **methylamine** ($\text{CH}_3\text{-NH}_2$).

Amides 1.5 marks: Functional group -C(=O)-NH_2 ; general formula R-C(=O)-NH_2 . Example: **ethanamide** ($\text{CH}_3\text{-C(=O)-NH}_2$).

OR

Catenation 1 mark: Carbon's self-linking ability to bond with other carbon atoms and form long chains, branched chains and rings is called **catenation**.

Contribution to diversity 1 mark: Because carbon can form chains of thousands of atoms with no apparent limit, the number of possible structures is enormous. No other element can compete with carbon in this regard (silicon can only form short chains).

Two other factors 1 mark: (1) **Isomerism** — same molecular formula can give many different structural arrangements; (2) **Functional groups** — replacing H atoms with different functional groups creates new classes of compounds with distinct properties.

SECTION C — Long Questions Answer Key (4 × 5 = 20 Marks)

3(i) Homologous series + general formulae applications 5 marks

(a) Definition and five characteristics 3 marks

Definition: A series of related compounds in which adjacent molecules differ by a $-\text{CH}_2-$ unit is called a homologous series.

Five general characteristics (from section 13.2.1):

1. Family members have the **same functional group**.
2. They have the **same general formula**.
3. Adjacent members differ by the unit $-\text{CH}_2-$.
4. Members display a **trend in physical properties** (e.g. boiling point increases with chain length).
5. Members possess **similar chemical properties** (same reactions because same functional group).

Award 1 mark for definition + 2 marks for at least four characteristics.

(b) Molecular formulae using general formulae 2 marks:

Required	General formula	n	Molecular formula
Alkane, 5 C	$\text{C}_n\text{H}_{2n+2}$	5	C_5H_{12} (pentane)
Alkene, 4 C	C_nH_{2n}	4	C_4H_8 (butene)
Alcohol, 3 C	$\text{C}_n\text{H}_{2n+1}\text{OH}$	3	$\text{C}_3\text{H}_7\text{OH}$ (propanol)
Carboxylic acid, 2 C	$\text{C}_n\text{H}_{2n+1}\text{COOH}$	1*	CH_3COOH (ethanoic acid)

*For carboxylic acid with 2 carbons total: use $n=1$ in $\text{C}_n\text{H}_{2n+1}\text{COOH}$, giving CH_3COOH .

OR

(a) Structural isomerism and catenation 3 marks

Catenation 1 mark: Carbon's ability to bond with other carbon atoms to form chains, branched chains and rings. No other element can form chains as long as carbon — silicon forms only short chains. This generates an enormous variety of structures.

Structural isomerism 1 mark: Compounds with the **same molecular formula but different structural arrangements** are structural isomers. A single molecular formula can produce many different compounds, each with different physical and chemical properties.

Combined effect 1 mark: Together, catenation (many chain lengths and shapes possible) and isomerism (multiple structures per formula) account for millions of organic compounds from relatively few elements.

(b) Structural isomers of C_4H_{10} 2 marks:

Isomer 1 – n-Butane (straight chain): $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3$ Isomer 2 – iso-Butane (branched chain): $\text{CH}_3\text{ | CH}_3\text{-CH-CH}_3$

Both have molecular formula C_4H_{10} ; n-butane has a straight carbon chain while iso-butane has a methyl branch on the second carbon.

3(ii) Saturated/unsaturated hydrocarbons; alkanes, alkenes, alkynes 5 marks

(a) Saturated and unsaturated hydrocarbons 3 marks

Hydrocarbons: compounds containing carbon and hydrogen only.

Saturated (alkanes): all C–C bonds are **single bonds**; general formula C_nH_{2n+2} . Example: methane (CH_4). Also called alkanes. Each C is bonded to four atoms. 1 mark

Unsaturated — two sub-types: 2 marks

- **Alkenes:** contain at least one C=C **double bond**; general formula C_nH_{2n} . Example: ethene ($CH_2=CH_2$).
- **Alkynes:** contain at least one C≡C **triple bond**; general formula C_nH_{2n-2} . Example: ethyne ($HC\equiv CH$).

(b) Structural and condensed formulas 2 marks :

(i) Ethene (C_2H_4): $\begin{array}{c} H & H \\ \backslash & / \\ C & = & C \\ / & \backslash \\ H & H \end{array}$ Condensed: $CH_2=CH_2$ (ii) Ethyne (C_2H_2): $H-C\equiv C-H$ Condensed: $HC\equiv CH$ (iii) Propene (C_3H_6): $\begin{array}{c} H & H & H \\ | & | & | \\ H-C & = & C-C-H \end{array}$ Condensed: $CH_3-CH=CH_2$ | H

OR

(a) Open-chain, cyclic, straight-chain, branched-chain 3 marks

Open-chain: carbon atoms are linked in a linear pattern — either straight or branched. Also called acyclic compounds. 1 mark

Cyclic: organic compounds containing rings of atoms are called cyclic or closed-chain compounds.

Examples: cyclopropane, benzene. 1 mark

Straight-chain: all carbons connected in one continuous chain (e.g. butane $CH_3CH_2CH_2CH_3$). **Branched-chain:** carbon atoms attached on the sides of the main chain.

iso-Butane: $CH_3-CH(CH_3)-CH_3$ (branch on C2) neo-Pentane: $CH_3-C(CH_3)_2-CH_3$ (two branches on C2)

1 mark

(b) Identification 2 marks (0.5 each):

- (i) $CH_3-CH_2-CH_2-CH_3$ — **straight-chain** (butane).
- (ii) $CH_3-CH(CH_3)-CH_3$ — **branched-chain** (iso-butane).
- (iii) Cyclopropane — **cyclic** (ring of 3 carbons).
- (iv) Benzene — **cyclic** (aromatic ring of 6 carbons).

3(iii) Identify functional groups in five given compounds 5 marks

Award 1 mark per compound for correct functional group + class + general formula (3 elements, so 1/3 each, but award the full mark if at least 2 of 3 elements are correct).

Compound	Functional group	Class	General formula
(a) CH_3OH	–OH (hydroxyl)	Alcohol	R–OH
(b) $\text{CH}_3\text{–O–CH}_3$	C–O–C (ether linkage)	Ether	R–O–R'
(c) CH_3CHO	–CHO (carbonyl + H)	Aldehyde	R–CHO
(d) CH_3COCH_3	–CO– (carbonyl between alkyls)	Ketone	R–CO–R'
(e) CH_3COOH	–COOH (carboxyl)	Carboxylic acid	R–COOH

Key distinctions to memorise:

- Alcohol (–OH on alkyl) vs Phenol (–OH on benzene ring) vs Ether (O between two alkyls).
- Aldehyde (–CHO: carbonyl carbon has one H) vs Ketone (–CO–: carbonyl carbon has no H, flanked by two alkyls).
- Carboxylic acid (–COOH) vs Ester (–COO–R'): ester has no –OH on the carbonyl carbon.

OR

(a) Functional group definition and importance 2 marks: An atom or group of atoms that gives a family of organic compounds its characteristic properties is called a functional group. Three reasons: (1) basis for naming; (2) tool for classifying into homologous series; (3) site of chemical reactivity.

(b) Identify class and functional group 3 marks (1 mark for any three correct):

Compound	Class	Functional group
(i) $\text{CH}_3\text{COOCH}_3$	Ester	–C(=O)–O– (–COO–)
(ii) CH_3NH_2	Amine	–NH_2
(iii) CH_3CONH_2	Amide	–C(=O)–NH_2
(iv) $\text{C}_6\text{H}_5\text{OH}$	Phenol	–OH on benzene ring
(v) $\text{CH}_3\text{CH}_2\text{I}$	Haloalkane	C–I (halogen)

3(iv) Alcohols, ethers, carboxylic acids, esters + isomer question 5 marks**(a) Four functional group classes** 4 marks (1 mark each):

Class	Functional group	General formula	Named example
Alcohol	–OH (hydroxyl)	R–OH	Ethanol (CH ₃ CH ₂ OH)
Ether	C–O–C (ether linkage)	R–O–R'	Dimethyl ether (CH ₃ –O–CH ₃)
Carboxylic acid	–COOH (carboxyl)	R–COOH	Ethanoic acid (CH ₃ COOH)
Ester	–C(=O)–O–	R–C(=O)–O–R'	Methyl ethanoate (CH ₃ COOCH ₃)

(b) Ethanol and dimethyl ether — isomers? 1 mark: Yes, they are **functional group isomers** (a type of structural isomers). Both have the molecular formula C₂H₆O but differ in the arrangement of atoms: ethanol has the –OH group attached to an ethyl group (it is an **alcohol**), while dimethyl ether has oxygen bridging two methyl groups (it is an **ether**). This is a textbook example in section 13.3.

OR**(a) Amines and amides** 2 marks:

Class	Functional group	General formula	Example
Amine	–NH ₂ (amino)	R–NH ₂	Methylamine (CH ₃ NH ₂)
Amide	–C(=O)–NH ₂	R–C(=O)–NH ₂	Ethanamide (CH ₃ CONH ₂)

(b) Classify four compounds 3 marks:

Compound	Class	Functional group
(i) CH ₃ NH ₂	Amine	–NH ₂
(ii) CH ₃ CONH ₂	Amide	–C(=O)–NH ₂
(iii) CH ₃ COOC ₂ H ₅	Ester	–C(=O)–O–
(iv) CH ₃ CH ₂ COOH	Carboxylic acid	–COOH